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Non-Solar UV Produced Ions Observed Optically From the "CRIT I" Critical Velocity Ionization Experiment

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A critical velocity ionization experiment was carried out with a heavily instrumented rocket launched from Wallops Island at dawn on May 13, 1986. Two neutral barium beams were created by explosive shaped charges released from the rocket and detonated at 48° to B at altitudes near 400 km and below the solar UV cutoff. Critical velocity ionization was expected to form a detectable ion jet along the release field line, but instead an ion cloud of fairly uniform intensity was observed stretching from the release field line across to where the neutral barium jet reached sunlight. The process creating these ions must have been present from the time of the release and the efficiency is estimated to be equivalent to an ionization time constant of 1800 s. This ionization is most likely from collisions between the neutral barium jet and the ambient atmospheric oxygen, and if so, the cross section for collisional ionization is 9×10^{-18} cm^2. A critical velocity ionization process may have been present during the first few tenths of a second after release, but its efficiency cannot have exceeded an equivalent ionization time constant of about 1800 s.

INTRODUCTION

During the last years a number of experiments have been performed both in the laboratory and in space to explore the concept of a critical ionization velocity (CIV) proposed by *Alfvén* [1954] and *Alfvén and Arrhenius* [1975] as part of a theory for the formation of planetary systems. Recent reviews have been given by *Newell* [1985], *Lai and Murad* [1989], *Torbert* [1988 and 1990], and *Piel* [1990]. The ionization takes place through collective plasma processes with the energy provided by the neutrals moving across a magnetic field embedded in the background plasma. For the ionization process to take place the velocity of the neutrals must exceed the critical ionization velocity determined by the kinetic energy in the velocity component perpendicular to the magnetic field being equal to the ionization potential of the neutrals.

In this paper we report on the results of the optical observations from a CIV experiment with two barium shaped charge releases carried out before dawn on May 13, 1986, from Wallops Island. The experiment was a continuation of earlier CIV experiments [*Wescott et al.*, 1986*a*,*b*], but with much improved diagnostics on the rocket and the ejected subpayload. The barium vapor was created below the solar terminator to prevent photoionization. The Alfvén critical velocity ionization process was expected to ionize a fraction of the neutral barium, and those ions would be observed as they rose up the field lines into the sunlight about 40 to 50 km above the detonations.

First, the general geometry of the releases is given based on the optical observations and the instrumentation on the rocket. The remaining part of the paper covers the analysis of the optical data and the search for evidence of CIV ionization. The experiment was designed with the expectation that ions would be created near the release by a relatively short lived critical velocity ionization process to form a field aligned jet along the release field line similar to the "Porcupine" experiment [*Haerendel*, 1982]. The "Porcupine" barium shaped charge experiment was carried out from Esrange, Sweden, on March 19, 1979, with injection about 100 km below the terminator and at a 28° angle to B. An approximately 15 km wide ion cloud was observed near the release field line. The source of these ions created below the terminator was attributed to a CIV process, and a recent reevaluation of the data indicates that the CIV ion cloud contained about 20% of the neutrals with velocities in excess of the critical ionization velocity [*G. Haerendel*, private communication, 1989].

However, in the experiment described here, no such ion jet was observed. Instead, an ion cloud of relatively uniform and low brightness, less than 100 R, was observed in the region above the solar terminator and between the release field line and the neutral barium jet. Since the neutral jet in this region was well below the terminator, these ions must have been produced by a process other than ionization by solar UV; further, the process must have been present continuously from the release.

To investigate the nature of this non-solar UV ion cloud, we present a computer simulation of the observations which shows that the number of ions produced followed:

$$N_i = N_{neutral} \times (1 - \exp^{-t/\tau})$$

with an ionization time constant, τ, of about 1800 s. The analysis can place an upper limit on the ion production rate from a CIV process in this experiment. Various sources for these ions are considered, and collisions between the neutral barium and the ambient atmosphere appears to be the most likely process, although a CIV process may be present near the release point. Assuming collisional ionization the analysis allows the ionization cross section to be calculated, which is an important piece of information for future release experiments.

OPTICAL OBSERVATIONS AND RELEASE GEOMETRY

The primary purpose for the optical observations was to obtain an inventory of both ions and neutrals in each barium injection and to obtain a differential velocity function of the CIV ions produced. To accomplish these goals three optical sites were located at Wallops Island, Virginia (37.861N, -75.457E), which is about 2 km from the launcher; Duck, North Carolina,

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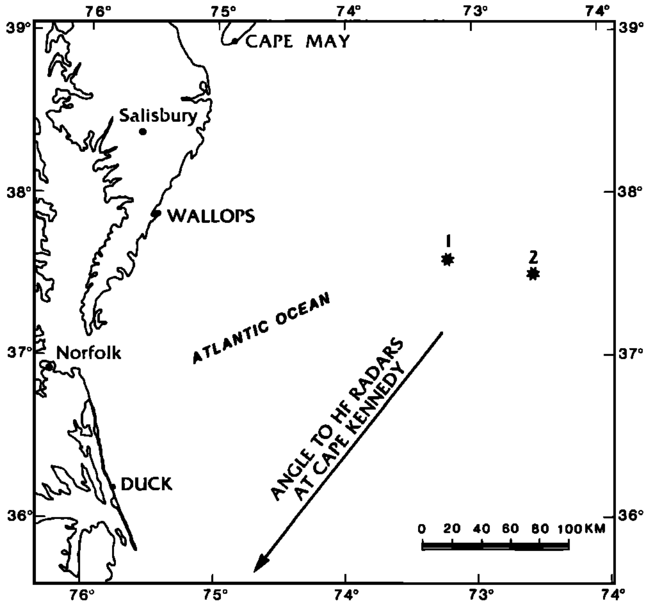

Fig. 1. Geographical positions of releases and main optical sites.

(36.182N, -75.751E); and Cape May, New Jersey (38.945N, -74.883E). At Wallops Island the University of Alaska and University College London operated 2 imaging photon detector (IPD) systems observing at the 5535Å neutral barium and at the 4554Å barium ion emission lines respectively. These lines are the dominant ion and neutral solar fluorescent emission lines. At Duck the University of Alaska operated an unfiltered intensified silicon intensifier target (ISIT) TV system and an intensified film camera observing in the 4554Å barium ion line. At Cape May the Max Planck Institute had a 1.8° field of view intensified secondary electron conduction (SEC) TV system filtered at 4554Å and 5535Å, a 5.6° field of view intensified change injection device (CID) TV system filtered at 4554Å, and a larger field of view unfiltered TV system. In addition, Technologies Incorporated operated an ISIT TV camera and film cameras at the Air Force Geophysics Laboratory, Bedford, Massachusetts (42.45N,71.27W). The location of the two barium releases and the three main optical sites are shown in Figure 1.

The experimental configuration chosen for the releases was detonation of the shaped charges upwards at an angle of 45° to the magnetic field direction and that the direction of the neutral jets be in the plane containing the magnetic field line and an HF radar site at Cape Kennedy and that the jet be directed away from the radar. The results of these requirements were neutral jets intended at an azimuth of 15° and an elevation of 64°.

One somewhat unfortunate consequence of this release geometry is that the neutral jet will go up, reaching sunlight at about the same time that the CIV ions would become visible. Figure 2 shows the look angles to the two releases from the three sites to illustrate this point. The heavy lines indicate the distance covered by neutrals and CIV ions at about 4 s when the neutral jet reaches the 19 km screening height above which the barium becomes optically detectable [*Stenbaek-Nielsen et al.*, 1984]. The screening height for the ionizing solar UV radiation (calculated below) is at about 34 km. When the neutral jet reaches this altitude it will ionize rapidly creating a bright ion cloud which might interfere with the observations of the CIV ions. To avoid interference, the intent was to keep this ion cloud outside the field of view of sensitive imagers, in particular the IPD observing in the ion line. A computer program using real time tracking radar data provided updates of look angles to the

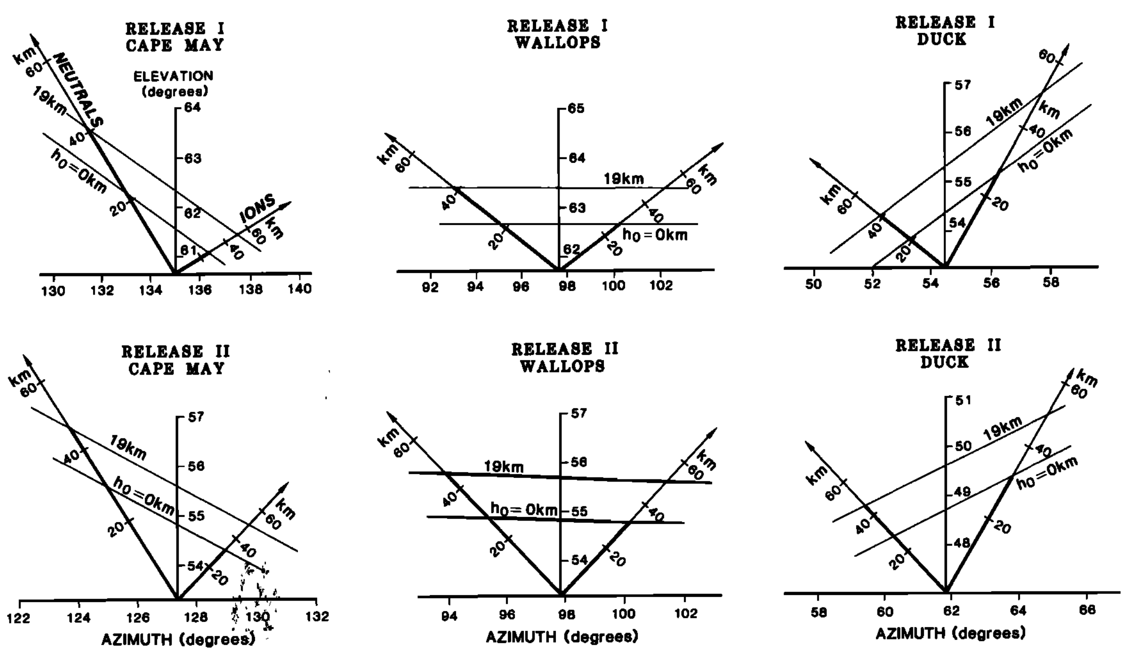

Fig. 2. Look angles to the two releases from the three optical sites. The neutral jet is going up towards the left and the magnetic field line to the right. The location of the geometrical terminator and the terminator corresponding to a screening height of 19 km above which the barium may be observed optically are shown. The heavy lines indicate that when the neutral jet reaches the 19 km screening height and becomes visible, ions created near the release will barely be at the geometric screening height.

releases. However, this was not entirely successful and only the ISIT TV cameras recorded both releases fully.

Various release parameters are listed in Table 1. These data are based on a best fit of the optical data and information from radar and telemetry. The magnetic field was calculated from the 1985 Internationl Geomagnetic Reference Field [IGRF85] model of the Earth's internal field updated to the epoch of the experiment. The direction of the neutral jet was derived by triangulation and the direction is estimated to be accurate to within about a degree. The main payload was located a few km in front of the release and very near the center axis of the neutral jet. The subpayload was above and outside the jet, but the edge of the jet at cone angles of 15.8° of and 11.5° respectively for the two releases, cut across the field line through the payload about 1 km from the release and ions created there would reach the payload.

We assessed the performance of the two shaped charges. Since the releases were below the terminator, the data are not ideal for this assessment, but there is nothing in the optical data which indicates that the yield and jet velocities differed between the two releases. Further, the velocity distribution function and yield appear to be similar to that observed for the SR90 release (E. M. Wescott et al., SR90, strontium shaped-charge critical ionization velocity experiment, submitted to *Journal of Geophysical Research*, 1989) and reproduced here as Figure 3. The angular

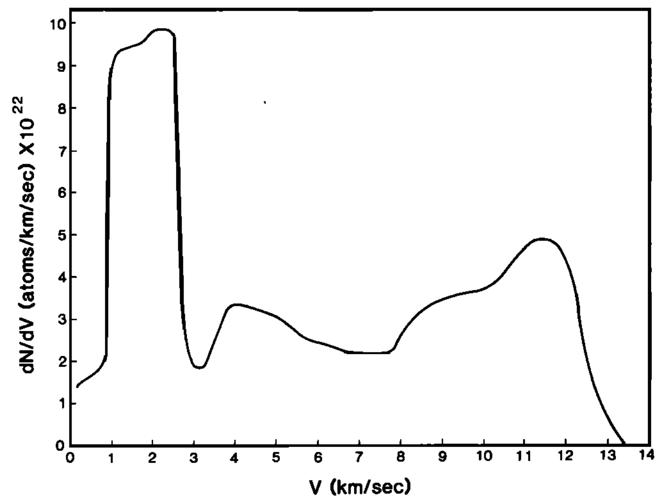

Fig. 3. Differential velocity distribution function as observed in the SR90 release (E. M. Wescott et al., SR90, strontium shaped-charge critical ionization velocity experiment, submitted to *Journal of Geophysical Research*, 1989). The ordinate scale is derived from the assumption that 15% of the barium liner is vaporized.

TABLE 1. Release Geometry for CRIT I Launched From Wallops Island on May 13, 1986

	Release 1	Release 2
Time (UT)	07:46:00	07:47:25
Latitude (N)	37.5983	37.5037
Longitude (E)	-73.2113	-72.5554
Altitude (km)	398.965	373.057
Magnetic field:		
B (nT)	43976.	44397.
Azimuth (°)	349.1	348.6
Elevation (°)	-66.8	-66.6
Direction to Sun:		
Azimuth (°)	44.9	45.8
Elevation (°)	-20.3	-19.8
Screening height (km)	-21.6	-25.5
Direction of neutral jet:		
Azimuth (°)	14.7	14.0
Elevation (°)	64.2	64.2
Direction of ion jet (-B):		
Azimuth (°)	169.1	168.6
Elevation(°)	66.8	66.6
Angle (°) between:		
Neutr. jet and ion jet	47.8	47.9
Neutr. jet and dir sun	87.7	87.5
Ion jet and dir Sun	148.2	149.1
Location of payloads:		
Main		
Latitude (N)	37.6051	37.5185
Longitude (E)	-73.2093	-72.5515
Altitude (km)	400.768	377.021
Dist. from burst (km)	1.987	4.344
Angle to jet (°)	1.2	1.7
Sub		
Latitude (N)	37.5962	37.5062
Longitude (E)	-73.2125	-72.5565
Altitude (km)	402.766	379.833
Dist. from burst (km)	3.811	6.783
Angle to jet (°)	29.7	23.4

Distances to sunlight are given in Figure 2.

distribution is nearly Gaussian with the half width, half maximum cone angle 6.6° for velocities above 7.0 km/s, 10-11° for velocities between 7.0 and 3.5 km/s, and 21-24° at the lowest velocities. The details of the distribution functions for the barium jet may differ from that of the SR90 release, but the observations of the major features (tip and peaks) in the barium jet are consistent with the SR90 release to better than a few tenths of km/s.

ATMOSPHERIC TRANSMISSION

For planning of the experiment and for data analysis it is important to know the brightness and the spectral content of the sunlight in the region of the releases. Above the geometric shadow height sunlight travels through varying parts of the atmosphere to the region of the release. The path may be characterized by the screening height defined as the height (km) of the lowest point on the ray path. The corresponding optical depth for a given wavelength can be calculated given the absorption cross section and an atmospheric model. For ionizing light (wavelengths shorter than 3266Å) the effects of ozone, which strongly absorbs UV radiation, must also be included. Using the atmospheric model given in the U.S. Standard Atmosphere Supplements, U.S. Government Printing Office [1966], and atmospheric extinction rates [*Guttman*, 1968] plus ozone densities and absorption cross sections given by *Banks and Kockarts* [1973], attenuation as function of screening height was calculated for the dominant barium neutral and ion emission lines at 5535 Å and 4554 Å respectively, and for ionizing light at wavelengths less than 3265 Å. It should be noted that the ozone content of the atmosphere is highly varying and the extent to which the chosen model reflects the actual conditions at the time of the releases is uncertain.

Figure 4 shows the calculated relative intensities of the ionizing UV radiation, the 4554 Å barium ion emission and the 5535 Å barium neutral emission versus screening height. Considering only the absorption due to ozone, the maximum UV screening occurs at 18 km. Below 18 km absorption by the regular air-

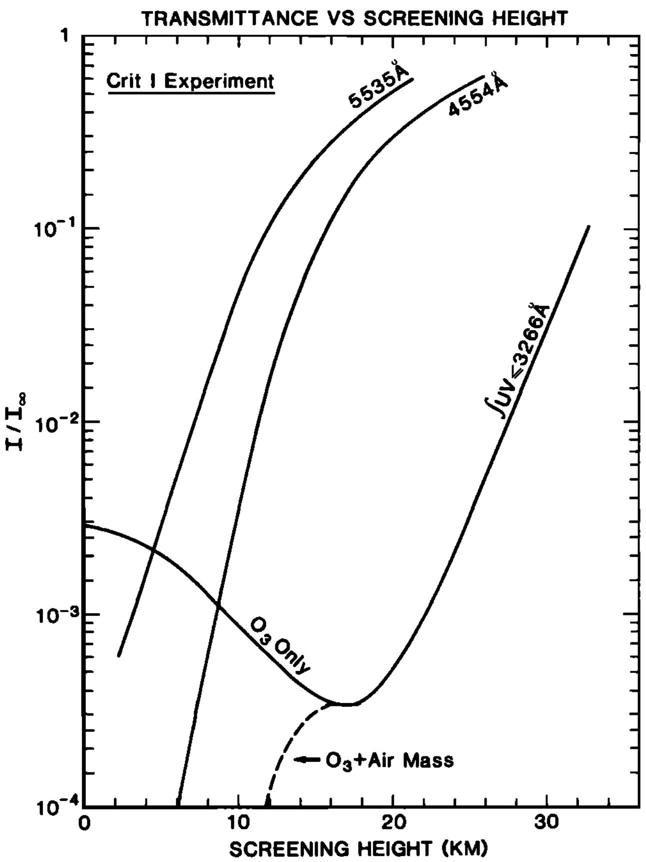

Fig. 4. Relative intensities of solar illumination at the releases at relevant wavelengths calculated as function of screening height.

mass dominates. The absorption of light at 4554Å and at 5535Å is much less than the absorption of the UV ionizing light. The limiting screening height depends on the brightness of the streak, but based on data from previous release experiments we can detect a barium streak above a screening height of about 19 km [*Stenbaek-Nielsen et al.*, 1984].

ISIT TV Observations

The ISIT TV system operated at Duck, North Carolina, observed both releases. It was operated unfiltered to obtain the best possible images from which the release position and the direction of the neutral jet and, if observable, the critical velocity ions. Even with the application of various image enhancing techniques there was no evidence of an ion streak along the burst field line.

The ISIT provided data with high time resolution throughout the experiment, and although the camera is less sensitive than the IPD which only provided data for 10 s on the first release, it was well pointed to observe ions along the release field line. An estimate of the sensitivity of the camera was made using the stars present in the data. The limiting star magnitude was about 8.0 which together with the spectral response function of the detector and the barium emission rates calculated by *Stenbaek-Nielsen* [1989] results in a minimum line of sight integrated (column) barium ion density of $5 \times 10^9/cm^2$. If the column density had been larger, the ion jet would have been observable in the ISIT TV camera images.

IPD Ion Observations

The primary data for the analysis were obtained by an imaging photon detector system observing in the dominant barium ion line at 4554Å and operated at a site near the launching complex at Wallops Island. The filter width was 28.8° FWHM with 60% peak transmission, sufficient to observe the emission line across the entire 10° field of view. The images were recorded digitally on a Compaq computer with 5 s integration time. The IPD processes the detected photons serially and the image is built up in computer memory. The upper limit on the number of photons that can be processed per second is near 100,000. At higher rates the detector may be damaged and an electronic switch turns the IPD off. We intended to position the cameras for optimum view of the release field line above the terminator. But real time update information about the trajectory was not received and the bright ion cloud created by photoionization when the neutral jet came up over the UV terminator appeared in the field of view. Two complete 5 s images were obtained before the detector turned off, but as luck would have it, the "wrong" pointing direction was almost perfect for the unexpected non-solar UV produced ion cloud that appeared.

The second release was observed from Wallops to take place embedded in a large diffuse ion cloud from the first release which effectively precluded the IPD from making useful observations.

The two IPD ion images obtained on the first release are of critical importance to the present study. The thermal emissions from the explosion were detected in the frame covering the release, but nothing else was seen in this frame, which covered the time period -2.0 s to +3.0 s from the release. In the next image, covering the time period 5.0 to 10.0 s from release, ions are present, but not as a well defined field aligned streak, as would have been expected from an efficient ionization process active during the first second or so after the release.

Since the burst frame did not show any barium ions it was used as representative of the sky background present and by subtracting that from the following image the background, including the stars, was eliminated. The resulting image, filtered to reduce the high frequency spatial noise, is shown in Figure 5.

To help in the interpretation of the image various lines have been inserted. The release location is identified by the cross at the bottom of the frame. The magnetic field line (IGRF85 updated to the epoch of the experiment) extends up to the right from the release point while the direction of the neutral jet is towards the upper left. The points along B and the neutral direction are at 10 km intervals. The top is delineated by the maximum distance a neutral or ion at 13.5 km/s, the tip of the jet, can reach in 10 s, and the bottom by the terminator at 19 km screening height.

The ions are in two parts: The very bright circular cloud in the left center part of the image is the cloud produced by photoionization and is of no interest to the present study. The second part is the faint cloud of fairly uniform intensity in the center of the image and within the shown delimiters. This cloud must have been produced from the neutral jet below the UV terminator. (The tendency of the ion cloud to be wider than indicated by the delimiters towards the left in the image is due to aspect: the ion cloud is viewed at an elevation angle of 67°, i.e. essentially from the underside. The delimiters are drawn in the plane of B and the jet, but the ions extend out to both sides of this plane because of the finite jet cone angle.)

The image was intensity calibrated based on the signal in the stars present and the measured characteristics of the interference

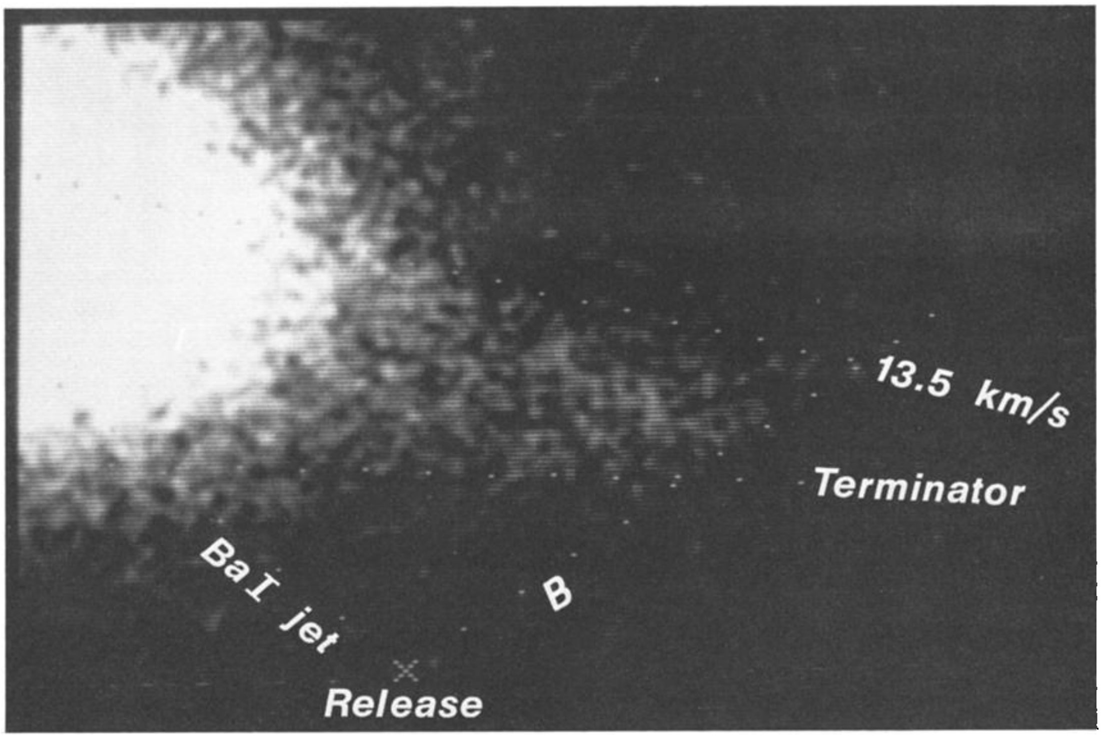

Fig. 5. IPD image of the first release observed in the 4554 Å barium ion line emission. The image is a 5 s exposure covering the time 5 to 10 s after release. The background has been subtracted and high frequency noise eliminated. The location of the release is at the bottom center of the image. The neutral jet was directed up towards the left and the direction of the magnetic field is up towards the right (marks at 10 km increments). The top delimiter is the maximum distance barium at 13.5 km/s can reach in 10 s, and the bottom is the terminator. The bright circular cloud dominating the image is from ions produced by solar UV illumination of the neutral jet. The faint cloud stretching right and slightly downwards from the bright ion cloud is from ions which must have been produced below the 19 km screening height terminator.

filter. Magnitudes and spectral types of the stars were obtained from the Smithsonian star catalog. Since the starlight and the emissions from the barium ion cloud has the same optical path to the detector, the use of the star background for calibration offers the advantage of automatically correcting for effects of atmospheric absorption and the possible presence of haze or thin clouds. The calibration of the image agrees well with the known characteristics of the detector and estimates of losses in the optics and in the atmosphere.

The cloud appears to be of relatively uniform intensity. It is doubtful that the apparent finer structure in the ion cloud has significance since there are only a few counts in each pixel. The average brightness of the brightest regions in the non-UV produced ion cloud is about 100 R, corresponding to a column density of 6×10^7 cm^{-2} if all ions are in full sunlight.

SYNTHETIC IMAGE CALCULATION

A computer program was written to simulate the observed IPD image. The program is in effect similar to that described in E. M. Wescott et al. (SR90, Strontium shaped-charge critical ionization velocity experiment, submitted to *Journal of Geophysical Research*, 1989) but employs a different computational path in order to include atmospheric absorption of the sunlight illuminating the ion cloud. This is an important part of the simulation since the lower altitude edge of the ion cloud is defined by the terminator. The program assumes no special physical process but only that the production of ions can be described by an ionization time constant. The effective ionization time constant was determined by matching the synthetic images with the observed image (Figure 5). To avoid potential problems with the illumination near the terminator the match was made near the top of the cloud, and the best match was found for an ionization time constant of 1800 s. The resulting synthetic image is shown in Figure 6 and is seen to give the essential feature of the observed image: A broad relatively uniform ion cloud stretching left from the injection field line.

The image shown in Figure 6 is an average of a number of synthetic images covering the 5.0 to 10.0 s time interval of the observed image. Each individual image is calculated for a given time using the actual release geometry and the same aspect geometry as the observed image. The velocity distribution function of the neutral barium was given in Figure 3; the angular distribution is Gaussian with a HWHM angle of 6.6° for the fast part of the jet (the slower material does not reach above the terminator before after +10 s). The ion cloud is assumed to be optically thin, and the emission rates for resonance of sunlight, given by *Stenbaek-Nielsen* [1989], include the velocity dependence due to the Doppler shift of the resonance lines relative to the deep Fraunhofer absorption lines in the solar spectrum at the main ion resonances. The emission rate is further adjusted for atmospheric absorption at 4554 Å as given in Figure 4. Finally, the effects of the barium ion gyromotion is neglected since it is small compared to the size of an image element.

The brightness of each picture element is calculated by numerical integration of the ion brightness along the line of sight. At each step in this integration, the ions would have been

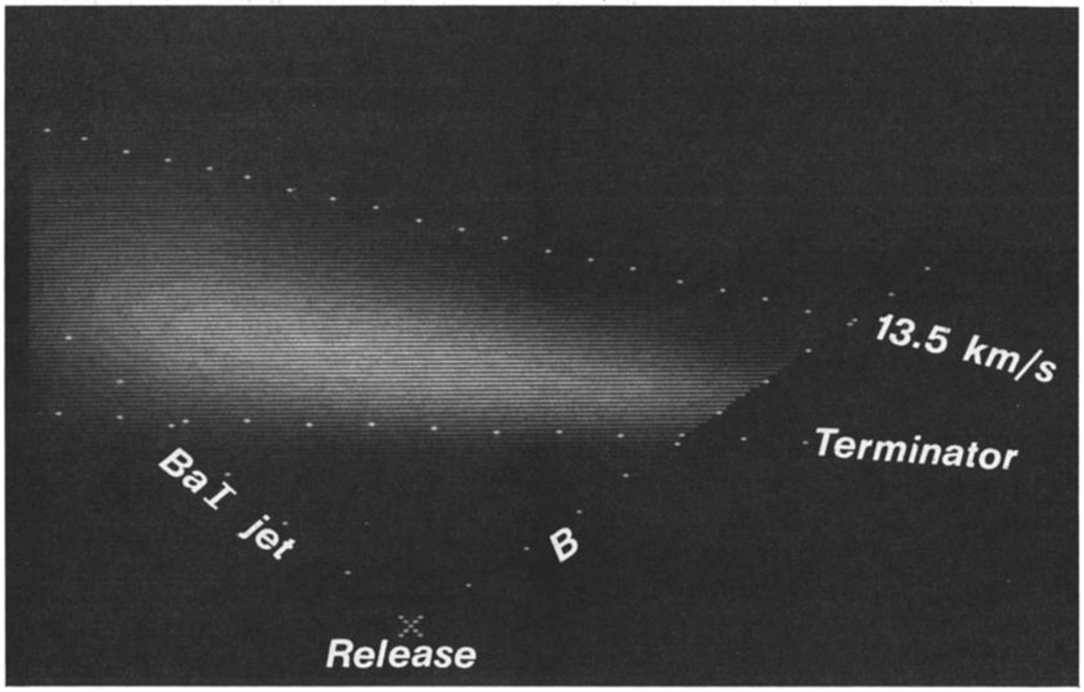

Fig. 6. Synthetic image calculated to simulate the observed IPD image in Figure 5. The 1800 s ionization time constant used in the simulation was selected to best match the observed intensity near the top of the ion cloud where full solar illumination can be assumed. The delimiters are the same as in Figure 5.

created along B and within the neutral beam below (all ions will go up given the injection geometry), and the total contribution to the brightness is evaluated through another numerical integration along B. The distances and angles involved from release to the point of ionization and then up along B to the line of sight, are defined, and for each selected point along B only atoms with one initial velocity will reach the line of sight at the time for which the image is calculated.

The model allows neutral and ion densities to be calculated at the location of the two payloads (neutral barium would only be present in significant amounts at the main payload—cf. Table 1). Figure 7 shows the results under two ionization scenarios both using the 1800 s ionization time constant determined from the optical observations. The solid line assumes ionization of all velocities. The dashed line represents the case of ionization due to collisions with the ambient atmosphere; in this case energy considerations require the neutral barium to have a velocity of at least 8.4 km/s relative to the ambient atmospheric oxygen atoms to ionize.

While the energy spectrum of the neutrals at any given time is a delta function, more velocities will be present in the ions because more than one path would lead to the position of the main payload. As a consequence, the ion densities are less structured than the neutral densities.

The model does not include the effects of a finite barium ion gyroradius. This is a reasonable assumption for the image calculation, but less so for the calculation of ion densities at the instrumented payloads located only a few km from the releases. The gyroradius of the fast jet is of the order of 200 m and the ions can therefore be displaced up to 400 m. This could affect the ion density considerably compared to the case where the ions remain on the field line of creation. The effect should be minor for the main payload since it is located near the center axis of the jet. However, the subpayload is encountering ions created near the edge of the neutral beam, and the calculated values would then be an underestimate.

SOURCES FOR ION PRODUCTION

Most barium releases rely on solar UV radiation for ionization. This is a fairly efficient process with an effective time constant of about 28 s [*Carlsten*, 1975; *Hallinan*, 1988] and thus much faster than the 1800 s estimated for the process in question here. (The difference in efficiencies is clearly illustrated by the relative brightness of the solar UV produced ion cloud in the upper left and the ion cloud in the center of Figure 5.) In the absence of direct sunlight there are several processes in addition to CIV ionization which can contribute to the ionization of the neutral barium jet: (1) photoionization by scattered UV light and (2) collisional ionization. There may also be ionization from the explosion process itself, but the temperature of the explosive products is substantially lower than the 5.21 eV required for ionization and more importantly, the explosive vapor cools as 1/time. An evaluation based on the Saha equation shows that any thermally produced ionization must be produced within the first few tens of microseconds in a very small volume at the release. Hence, the ionization would be expected to be confined to near the release field line which contrasts with the observed uniform ionization extending tens of km beyond with no discernible enhancement near the release field line. Consequently, we can disregard this process as contributing to the observed ionization.

Photoionization

Photoionization requires the presence of light of wavelengths less than 3265 Å. The primary source appears to be sunlight reaching the barium jet through scattering in the atmosphere.

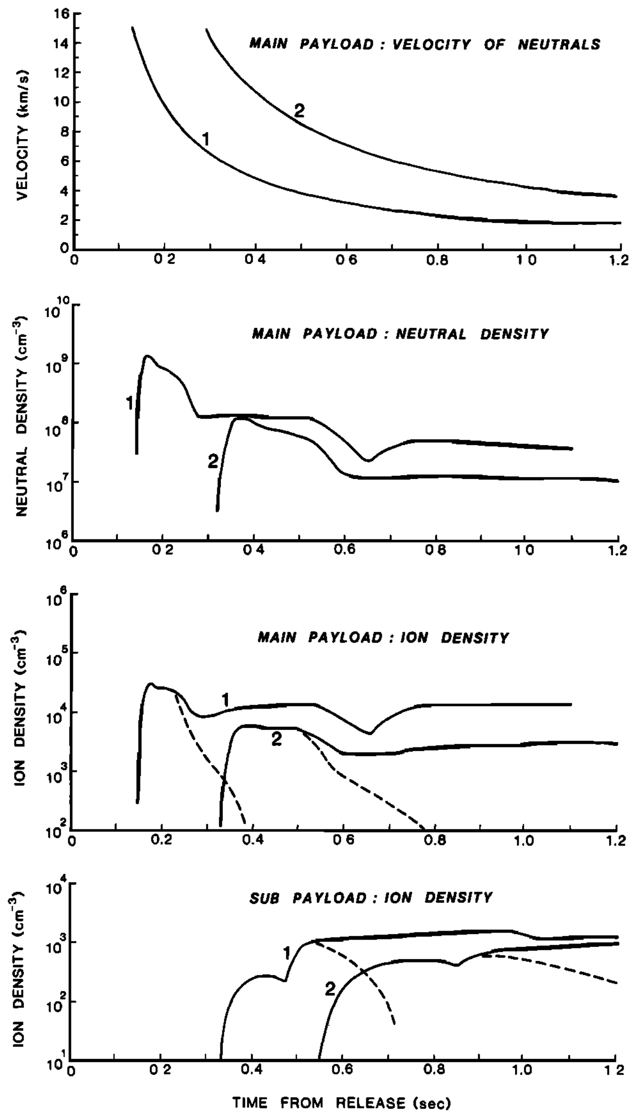

Fig. 7. Barium release parameters calculated at the positions of the two instrumented payloads. The top panel shows the velocity of the neutral barium passing the main payload as function of time after release. The center panels show the corresponding barium neutral and ion densities, and the bottom panel shows the ion density at the subpayload. The solid line gives the ion densities if we assume a constant ionization rate independent of velocity. If the ions are produced by collisions with ionospheric oxygen, there is an energy threshold at 8.4 km/s, below which no ionization occur. The resulting densities are given by the dashed line.

The moon was not visible from the release region and therefore need not be considered. Light from stars and the interstellar background, thermal radiation from the atmosphere and the Earth, and radiation from chemical processes in the atmosphere are estimated to be insignificant.

To estimate the flux of UV photons from scattered sunlight, we assume the scattering process to be Rayleigh scattering, that singly scattered photons dominate at the release, that the atmosphere above the UV terminator is optically thin and below optically thick, and that refraction can be neglected. Thus the problem is reduced to calculate the flux scattered towards the release from all parts of the sunlit atmosphere visible from the release.

For a given direction from the release the brightness in Rayleighe of the sky is

$$(4\pi \times 10^{-6})((3/16\pi)(1+\cos(\theta)) \times \int n(h)dl \times \int I(\lambda) \times \sigma(\lambda)d\lambda$$

In this equation the terms in front of the integrals are the scaling factor for the unit Rayleigh (4π to give equivalent isotropic flux) and the normalized angular scattering function for Rayleigh scattering (in units per steradian). The first integral is the column density (cm^{-2}) of the sunlit atmosphere along the line of sight; for an evaluation we used the model given in the U.S. Standard Atmosphere Supplements, U.S. Government Printing Office [1966]. The last integral gives the total number of photons scattered per atmospheric nucleus; using the solar flux given by *Banks and Kockarts* [1973] and the Rayleigh scattering cross sections given by *Bates* [1984] its value for wavelengths less than 3266Å is 3.5×10^{-10} photons/s.

A computer program was written to perform the detailed calculation using the actual release geometry. At the point on the UV horizon closest to the Sun we find a sky brightness of 2400 MR. The brightness falls rapidly off above this point reflecting the rapidly decreasing column density and is down by three orders of magnitude in only 1.5° of elevation. The decrease is less dramatic along the horizon where the same decrease in brightness requires 60° in azimuth.

The total omnidirectional UV photon flux at the release is 1.2×10^{13} photons cm^{-2} s^{-1} or only 0.02% of the full solar UV flux. The ionization time constant for this flux would be about 120,000 s, wholly insufficient to explain the observed ions, and it is unlikely that a more refined calculation would alter this conclusion.

Collisional Ionization

The neutral barium is streaming in the background atmosphere and some ionization may occur due to collisions. Two types of ionization processes can be envisioned: charge exchanges with ambient ions and collisions with ambient neutrals.

Charge exchange with ambient ions has a cross section of 2×10^{-16} cm^2 [*R. Torbert*, private communication, 1989]. The ambient ion density at the releases, as measured by the Millstone incoherent radar, was 6×10^4 cm^{-3} which results in an ionization rate of 1.4×10^{-5} s^{-1} corresponding to a time constant of 70,000 s. Thus, the contribution from this process towards the observed ionization would be insignificant.

For collisional ionization (or stripping) to be possible the barium must have an energy in excess of 5.21 eV (the ionization energy for barium) in the center of mass reference frame. At the altitude of the release the dominant constituent is atomic oxygen which, assuming the oxygen at rest, leads to a minimum velocity of the barium of 8.4 km/s. Therefore only the fastest part of the neutral jet will be able to ionize by atmospheric collisions.

Assuming the density of the neutral barium beam to be much less than that of the ambient atmosphere, the ionization rate per neutral barium atom for a given cross section, σ, is

$$I = n(h) \times \sigma \times v$$

where n(h) is the atmospheric density and v is the velocity of the barium atom relative to the target. At the release altitude of 400 km MSIS-86 [*Hedin*, 1987] gives $n = 5 \times 10^7/cm^3$ and with v = 12 km/s, the velocity of the peak in the velocity distribution function, and an ionization rate of 1/1800 per second, the ionization cross sections would be 9×10^{-18} cm^2.

We are not aware of any laboratory data with which a direct comparison can be made. *Firsov* [1959] made a theoretical analysis which for the velocity range of interest here, 8-13 km/s, yields cross sections in the range $5\times10^{-17}-8\times10^{-17}$ cm^2. This is considerably above our cross section of 9×10^{-18} cm^2. *Fleischmann et al.* [1972] compared Firsov's formula to more than 100 neutral-neutral ionization experiments including Ba, but all at energies several orders of magnitude above the beam energy in our experiment. They derived a semiempirical formula claimed to fit 80% of the cross sections to within a factor of about 2. Applying this formula we obtain cross sections in the range 2×10^{-18} to 3.5×10^{-18} cm^2 for barium neutrals between 8 and 13 km/s. Our cross section of 9×10^{-18} cm^2 is 2.5 times the Fleischmann et al. cross section for 13 km/s barium neutrals. Both Firsov and Fleischmann et al. concentrated on energies considerably above threshold which is clearly demonstrated by the fact that both formulas give finite cross sections below a barium neutral velocity of 8.4 km/s where the ionization process is energetically impossible. On the other hand, from measurements by *Maier and Murad* [1971] of stripping of molecular nitrogen near the energy threshold, *Lai and Murad* [1989] estimate that the cross section may be as large as 10^{-16} cm^2, which is 10 times the cross section we have inferred.

Our cross section is based on the 1800 s ionization time constant determined by matching the simulated and observed image. To avoid potential problems with atmospheric absorption the match was made near the top of the image farthest above the terminator. This part of the image would be from the fastest part of the neutral jet, and therefore, the cross section must be associated with velocities near 13 km/s. For velocities near threshold the cross section should fall off reaching 0 at 8.4 km/s. The effect in the image (in which ions from neutrals with velocities near 8.4 km/s are just appearing above the terminator) would be that the part of the ion cloud near the terminator would be less bright than predicted by the simulation. Although the observed image tends to support this, the calculation of the terminator in the model is at present too uncertain to warrant any definite claim. In Figure 7 the effect would be that the densities at the payloads would fall off more rapidly than the dashed line calculated using a constant ionization rate for all velocities above 8.4. km/s.

CONCLUSIONS

The optical observations on the first of the two releases revealed an ion cloud stretching from the release field line to the neutral jet (Figure 5). These ions were produced below the solar terminator and therefore must have resulted from an ionizing process other that direct photoionization by solar UV radiation. The process must have been present continuously from the time of release. A computer simulation of the image (Figure 6) has shown that the efficiency must have been nearly constant in time and equivalent to an ionization time constant of about 1800 s.

The probable source of this ionization is collisions with the ambient atmospheric oxygen. the observed image is almost exclusively due to ions with initial neutral velocities above the threshold of 8.4 km/s for collisional ionization, and the cross section required to match the observations is of order 10^{-17} cm^2, which is not an entirely unreasonable cross section. Model calculations [*Torbert*, 1988] indicate that the CIV effect should produce most of the ionization within 5 km from the release. A definite distance limit is set by excitation of ambient neutral oxygen, which will cool the electrons and damp the CIV effect when the barium density falls below 1/8 of the neutral oxygen density [*Newell and Torbert*, 1985]. In CRIT I, this would happen at distances exceeding 11 km from the explosion, while the observed ionization extended to at least 40 km.

Assuming that the source of the ionization is collisions with the ambient atmosphere, it is clear that this constitutes an important source that must be taken into account in future release experiments, notably the upcoming ionospheric releases from the Combined Release and Radiation Effects Satellite (CRRES), which will take place at an orbital velocity near 10 km/s. It would therefore be important to acquire more information about the ionization cross section near threshold in the 8 to 12 km/s velocity range.

When estimating the collision cross section it was implicitly assumed that the beam density is much lower than that of the ambient atmosphere. This is not the case immediately after release. Using the distribution function inferred from the SR90 release and an atmospheric density of 5×10^7 cm^{-3} [*Hedin*, 1987] the maximum beam density along the axis of the neutral jet will not be below the atmospheric density before 6 km, or 0.5 s, from the release. There is in the observed image a clear indication that the ions start to form essentially at the time of release (Figure 5) leaving open the possibility that collisional ionization only becomes important at a later time and that the ions observed near the release could be critical velocity ionization with similar efficiency. We must therefore conclude, that if a CIV process is present immediately after release, its ionization time constant cannot be smaller than 1800 s corresponding to a maximum ionization rate of 1/1800 or 0.06% per second.

Lai et al. [1988] have argued, based on observations made from the space shuttle, that CIV is less likely to occur as the ambient density decreases and they give an altitude limit of about 240 km. Although this might agree with the observations reported here, it would not explain the ionization observed in the "Porcupine" experiment at an altitude of 450 km. A comparison of the many CIV rocket experiments carried out by our team have now led us to believe that the critical parameter for CIV is the background electron density. In "Porcupine" the background electron density was 2×10^5 cm^{-3} [*Torbert*, 1988], much higher than in any of the other experiments conducted until CRIT II, carried out May 4, 1989, from Wallops Island. In this experiment the ambient electron density was 6×10^5 cm^{-3}, and preliminary analysis of the observational data indicates that, indeed, CIV took place [*Torbert*, 1989; *Stenbaek-Nielsen et al.*, 1989].

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